

# PRODUCT SPECIFICATION

# 产品规范

适用于 For  
K807 GNSS 板卡  
K807 GNSS Board



## INTRODUCTION

REVISION HISTORY / 修订历史

| Revision/版本 | Modification/更改                    | Date/日期    |
|-------------|------------------------------------|------------|
| 1.0         | New Release / 新发                   | 2023/09/25 |
| 1.1         | Modify the physical picture /修改实物图 | 2024/08/08 |

## DIRETORY / 目录

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# 1. Introduction / 简介

K807 GNSS is a full-constellation and multi-frequency positioning board developed by ComNav Technology based on Quantum III Soc with fully independent intellectual property rights. It supports BDS-2, BDS-3, GPS, GLONASS, Galileo, QZSS, IRNSS and other satellite system signals, with high-precision timing, ionospheric scintillation (S4, PHI), water vapor (PWV, ZTD) and other functions, on-board 8GB memory, it is suitable for surveying and mapping, foundation enhancement, high-end base stations and other fields.

K807 GNSS是司南导航基于具有完全自主知识产权的Quantum III开发的全系统全频点定位板卡，支持BDS-2、BDS-3、GPS、GLONASS、Galileo、QZSS、IRNSS等卫星系统信号跟踪，具备高精度授时，电离层闪烁(S4、PHI)，水汽(PWV、ZTD)功能，板载8GB内存，适用于测量测绘、地基增强、高端基站等领域。

## 1.1. Product Characteristics / 产品特性

Table 1. Product Characteristics / 产品特性

| Characteristics                           | K807                                                             |                                         |
|-------------------------------------------|------------------------------------------------------------------|-----------------------------------------|
| Signals<br>信号                             | Positioning<br>定位                                                | GPS: L1C/A, L2P, L2C, L5, L1C           |
|                                           |                                                                  | BDS-2: B1I, B2I, B3I                    |
|                                           |                                                                  | BDS-3: B1I, B3I, B1C, B2a, B2b          |
|                                           |                                                                  | GLONASS: G1, G2, G3*                    |
|                                           |                                                                  | Galileo: E1, E5b, E5a, E5 AltBoC*, E6c* |
|                                           |                                                                  | QZSS*: L1C/A, L2C, L5, L1C              |
|                                           |                                                                  | SBAS*: L1C/A, L5                        |
|                                           |                                                                  | IRNSS*: L5                              |
|                                           | Items with * will be adjusted with the version.<br>带*项会随同版本进行调整。 |                                         |
| Ionospheric Flicker Monitoring<br>电离层闪烁监测 | Signals<br>信号                                                    | GPS L1C/A, L2C, L2P, L5, L1C            |
|                                           |                                                                  | BDS-2: B1I, B3I                         |
|                                           |                                                                  | BDS-3: B1I, B3I, B1C, B2a, B2b          |

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|                             |                                     |                                                                            |                                                                                                 |
|-----------------------------|-------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                             |                                     |                                                                            | GLONASS: G1, G2                                                                                 |
|                             |                                     |                                                                            | Galileo: E1, E5b, E5a, E5 AltBoC, E6c                                                           |
|                             | Indicators<br>指数                    | Amplitude<br>Flicker Index<br>Monitoring<br>Accuracy<br><br>幅度闪烁指数监测<br>精度 | $\leq 0.1$ (Satellite Elevation Angle is higher than $15^\circ$ , 卫星仰角高于 $15^\circ$ )           |
|                             |                                     | Phase Flicker<br>Index<br>Monitoring<br>Accuracy<br><br>相位闪烁指数监测<br>精度     | $\leq 0.1\text{rad}$ (Satellite Elevation Angle is higher than $15^\circ$ , 卫星仰角高于 $15^\circ$ ) |
|                             |                                     | Observation<br>Frequency of<br>Flicker Index<br><br>闪烁指数观测频度               | $\geq 1/\text{min}$                                                                             |
| Water Vapor<br>水汽           | ZTD Measurement Error<br>ZTD测量误差    |                                                                            | $\leq 21\text{mm}$ (A Weather Meter needs to Be connected, 需要连接气象仪)                             |
|                             | PWV Measurement Error<br>PWV测量误差    |                                                                            | $\leq 3\text{mm}$ (A Weather Meter needs to Be connected, 需要连接气象仪)                              |
| Time to First Fix<br>首次定位时间 | Cold Start<br>冷启动                   |                                                                            | $< 20\text{s}$ (Adding Acceleration Capture Module, 增加捕获加速模块)                                   |
|                             | Hot Start (with RTC)<br>热启动 (使用RTC) |                                                                            | $< 10\text{s}$ (Typical, 典型)                                                                    |
| Signal Capture              | Reacquisition                       |                                                                            | $< 1\text{s}$                                                                                   |

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|                                |                                                                                               |                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 信号捕获                           | 失锁重捕                                                                                          |                                                                                                                         |
|                                | Signals Tracking Sensitivity<br>信号跟踪灵敏度                                                       | -155dBm                                                                                                                 |
|                                | Signal Capture Sensitivity<br>信号捕获灵敏度                                                         | -138dBm                                                                                                                 |
| Measurement Precision<br>测量准确度 | Pseudo-range Precision<br>伪距精度                                                                | ≤10cm                                                                                                                   |
|                                | Carrier Phase Precision<br>载波相位精度                                                             | ≤1mm                                                                                                                    |
| Accuracy<br>精度                 | PPS (RMS)<br>授时精度                                                                             | 7ns                                                                                                                     |
|                                | SPP Accuracy<br>标准单点定位精度                                                                      | H≤1.5m, V≤3m (1σ, PDOP≤4)                                                                                               |
|                                | Static Differential Accuracy (Supported by Compass Solution)<br>静态差分精度 (Compass Solution软件支持) | H:±(2.5+1×10 <sup>-6</sup> ×D)mm<br>V:±(5.0+1×10 <sup>-6</sup> ×D)mm<br>D为基线长度 (单位: mm)<br>D-Baseline length (Unit: mm) |
|                                | Speed Accuracy<br>测速精度                                                                        | ≤0.02m/s (1σ, PDOP≤4)                                                                                                   |
| PPP                            | PPP Convergence Time<br>PPP初始化时间                                                              | <15min                                                                                                                  |
|                                | PPP Accuracy<br>PPP精度                                                                         | H≤10cm, V≤20cm                                                                                                          |

|                          |                                                                                                     |                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| RTK                      | RTK Initialization Time<br>RTK初始化时间                                                                 | <10s (D<10km)                                                                                                   |
|                          | Initialization Reliability<br>初始化置信度                                                                | >99.9%                                                                                                          |
|                          | RTK Accuracy<br>RTK精度                                                                               | H:±(8+10 <sup>-6</sup> ×D)mm<br>V:±(15+10 <sup>-6</sup> ×D)mm<br>D为基线长度(单位: mm)<br>D-Baseline length (Unit: mm) |
| Anti-interference<br>抗干扰 | K807 is built-in narrowband and continuous wave suppression algorithm engine<br>K807 内置窄带和连续波抑制算法引擎 |                                                                                                                 |
| Data Rates<br>数据速率       | Measurements & Position<br>测量&定位                                                                    | Max 20Hz (Optional, 选配项)                                                                                        |
|                          | RTK: Positioning<br>RTK: 定位                                                                         | Max 20Hz (Optional, 选配项)                                                                                        |
| Data Formats<br>输出数据格式   | NMEA-0183                                                                                           | GPGGA, GPGSV, GPGLL, GPGSA, GPGST, GPHDT, GPRMC, GPVTG, GPZDA etc.                                              |
|                          | ComNav Binary (CNB)<br>司南二进制格式                                                                      | ComNav Self-Defined Binary<br>司南自定义二进制                                                                          |
|                          | CMR(GPS)                                                                                            | CMROBS, CMRREF                                                                                                  |
|                          | RTCM2.X                                                                                             | RTCM1, RTCM3, RTCM31                                                                                            |
|                          | RTCM3.X                                                                                             | 1004~1008,1012,1019,1020,1033,1042,1045/1046, 1230<br>MSM3~MSM7:1073~1077,1083~1087,1123~1127,1093~1097         |

|                            |                               |                                               |
|----------------------------|-------------------------------|-----------------------------------------------|
| Electrical<br>电气特性         | Storage<br>板载存储               | 8GB                                           |
|                            | Voltage<br>供电电压               | + 3.3V~3.45V DC                               |
|                            | Power Consumption<br>功耗       | 1.8W                                          |
| Environmental<br>环境要求      | Operating Temperature<br>工作温度 | -40°C~+85°C                                   |
|                            | Storage Temperature<br>储存温度   | -55°C~+95°C                                   |
| Antenna Interface<br>天线接口  | Impedance Matching<br>阻抗匹配    | 50Ω                                           |
|                            | Antenna Power<br>天线供电电压       | External 外部供电:<br>+3.3V~+5V @ (0-100) mA      |
|                            | Antenna Gain<br>天线增益          | 20~35dB                                       |
| Hardware Interface<br>硬件接口 |                               | UARTx2 , UART/USBx1 , PPSx1 , EVENTx2 , SPIx1 |
| Physical<br>物理参数           | Size<br>尺寸                    | 45.7mm×71.1mm×9.7mm                           |
|                            | Weight<br>重量                  | 38.4g                                         |
|                            | Package<br>封装                 | 28PIN                                         |



## 2. Size / 尺寸

In this section, product photo, three-side views and the dimension of K807 is provided for customers' further hardware design and installation.

本节提供了K807的实物图，三视图和对应的物理尺寸，便于用户进一步系统硬件设计和安装。



Figure 1. K807 Product Photo / K807 实物图

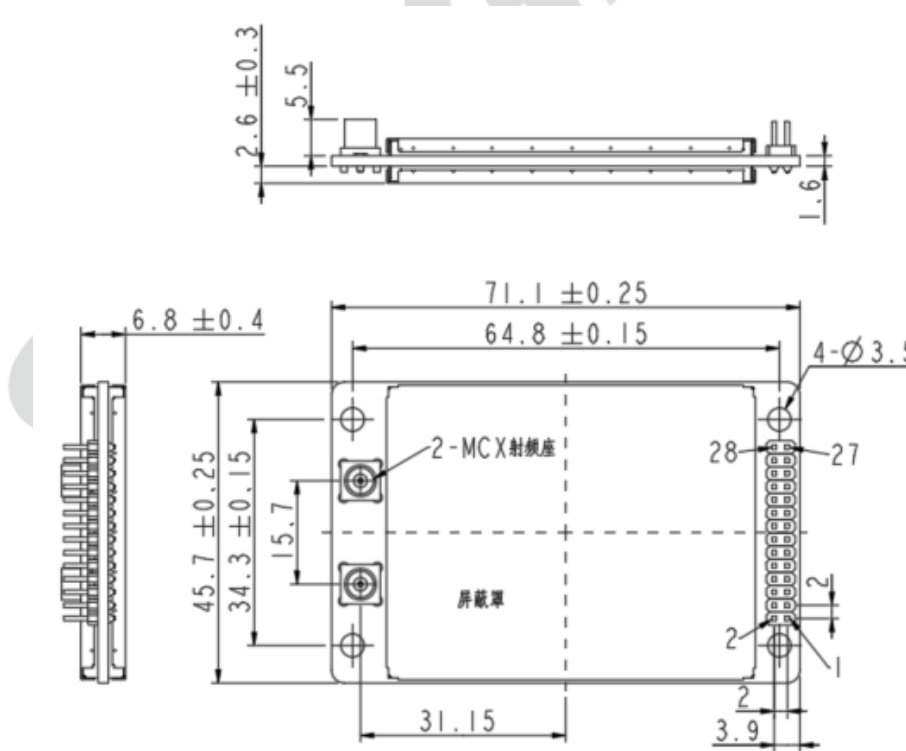


Figure 2. K807 Dimension View / K807 三视图

### 3. Pin Arrangement and Definition / 针脚标识与定义

K807 GNSS board includes 28Pin.

K807GNSS板卡包括28Pin。

|    |                |                |    |
|----|----------------|----------------|----|
| 1  | RFU2           | RFU1           | 2  |
| 3  | SPI0_CLK       | SPI0_CS        | 4  |
| 5  | LNA_PWR        | VIN            | 6  |
| 7  | USBA_PN        | COM3_RX/USB_PP | 8  |
| 9  | RST_SYS        | VARF           | 10 |
| 11 | EVENT2         | LED_RTK        | 12 |
| 13 | COM3_TX/EVENT1 | GND            | 14 |
| 15 | COM1_TX        | COM1_RX        | 16 |
| 17 | GND            | COM2_TX        | 18 |
| 19 | COM2_RX        | GND            | 20 |
| 21 | PV             | GND            | 22 |
| 23 | PPS            | LED_SAT        | 24 |
| 25 | SPI0_MISO      | SPI0_MOSI      | 26 |
| 27 | B_CODE         | TOD            | 28 |

Figure 3. K807 Includes 28 Pin / K807 包括28Pin

Table 2. Pin Definition of K807 28 Pin / K807 28Pin针脚定义

| Pin | Signal   | Type       | Description                           |            |
|-----|----------|------------|---------------------------------------|------------|
| 1   | RFU1     | /          | Reserve (Float)                       | 保留管脚（悬空）   |
| 2   | RFU2     | /          | Reserve (Float)                       | 保留管脚（悬空）   |
| 3   | SPI0_CLK | I/O        | SPI0 CLK                              | SPI1总线时钟信号 |
| 4   | SPI0_CS  | I/O        | SPI0 CS                               | SPI1总线片选信号 |
| 5   | LNA_PWR  | PWR<br>(I) | Power Supply for External Antenna LNA | 天线供电电源     |
| 6   | VIN      | PWR        | DC Power Supply for Card              | 板卡供电电源     |
| 7   | USBA_PN  | I/O        | USB Interface Data (-)                | USB数据信号(-) |

| Pin                                                              | Signal         | Type        | Description                                                              |                                      |
|------------------------------------------------------------------|----------------|-------------|--------------------------------------------------------------------------|--------------------------------------|
| 8                                                                | COM3_RX/USB_PP | Multiplexed | Received Data for COM3 Input /USB Interface Data (+)<br>Default: COM3_RX | 串口3输入信号/USB数据信号(+)<br>默认状态: 串口3输入信号  |
| 9                                                                | RST_SYS        | I           | System Reset                                                             | 系统复位信号                               |
| 10                                                               | VARF           | O           | Square Wave Output                                                       | 方波输出                                 |
| 11                                                               | EVENT2         | I           | Event2 Mark                                                              | 外部事件2输入                              |
| 12                                                               | LED_RTK        | O           | RTK Data LED Indicator                                                   | 在接收或发送差分信号时会输出高电平脉冲, 其他状态输出低电平       |
| 13                                                               | COM3_TX/EVENT1 | Multiplexed | Transmitted Data for COM3 Output /Event1 Mark<br>Default: UART3 TX       | 串口3输出信号/外部事件1输入<br>默认状态: 串口3输出信号     |
| 14                                                               | GND            | GND         | Ground Reference                                                         | 参考地                                  |
| 15                                                               | COM1_TX        | O           | Transmitted Data for COM1 Output                                         | 串口1输出信号                              |
| 16                                                               | COM1_RX        | I           | Received Data for COM1 Input                                             | 串口1输入信号                              |
| 17                                                               | GND            | GND         | Ground Reference                                                         | 参考地                                  |
| 18                                                               | COM2_TX        | O           | Transmitted Data for COM 2 output                                        | 串口2输出信号                              |
| 19                                                               | COM2_RX        | I           | Received Data for COM 2 input                                            | 串口2输入信号                              |
| 20                                                               | GND            | GND         | Ground Reference                                                         | 参考地                                  |
| 21                                                               | PV             | O           | Output Indicates Valid GPS Position when High                            | 位置有效指示信号                             |
| 22                                                               | GND            | GND         | Ground Reference                                                         | 参考地                                  |
| 23                                                               | PPS            | O           | Pulse Output Synchronized to OEM Time                                    | 同步卫星时间脉冲                             |
| 24                                                               | LED_SAT        | O           | Tracked Satellite Number Indicator                                       | 搜星时, 模块搜星数与高电平脉冲数一致; 不搜星时, 5S输出一次高电平 |
| 25                                                               | SPI0_MISO      | I/O         | SPI0 MISO                                                                | SPI1主输入从输出信号                         |
| 26                                                               | SPI0_MOSI      | I/O         | SPI0 MOSI                                                                | SPI1主输出从输入信号                         |
| 27                                                               | B_CODE         | O           | B_CODE                                                                   | B码授时                                 |
| 28                                                               | TOD            | O           | TOD                                                                      |                                      |
| Items with * will be adjusted with the version.<br>带*项会随同版本进行调整。 |                |             |                                                                          |                                      |

### 3.1. Remarks / 说明

#### 1. Electrical Characteristics / 电气特性

COM1/2/3(TX&RX), SPI, LED\_STAT/RTK, PV, RTS\_SYS, PPS, EVENT1 and EVENT2 are LVCMOS 3.3V level, All these signals are LVCMOS/LVTTL 3.3V compatible.

COM1/2/3 (TX&RX) , SPI, LED\_STAT/RTK, PV, RTS\_SYS, PPS, EVENT1和EVENT2为LVCMOS 3.3V 电

平，所有这些信号均兼容LVCMOS / LVTTL 3.3V。

Table 3. LVCMOS 3.3V Electrical Standard / LVCMOS 3.3V电气标准

| Symbols 符号      | Description 描述                       | Min 最小   | Max 最大   |
|-----------------|--------------------------------------|----------|----------|
| V <sub>IH</sub> | Input high voltage<br>输入高电压          | 2.0V     | VCC+0.3V |
| V <sub>IL</sub> | Input low voltage<br>输入低电压           | -0.3V    | 0.8V     |
| V <sub>OH</sub> | High-level output voltage<br>高电平输出电压 | VCC-0.4V | --       |
| V <sub>OL</sub> | Low-level output voltage<br>低电平输出电压  | --       | 0.41V    |
| I <sub>OH</sub> | Sourcing current<br>拉电流              |          | 8mA      |
| I <sub>OL</sub> | Sinking current<br>灌电流               |          | 8mA      |

Table 4. LVTTL 3.3V Electrical Standard / LVTTL 3.3V电气标准

| Symbols 符号      | Description 描述                       | Min 最小   | Max 最大   |
|-----------------|--------------------------------------|----------|----------|
| V <sub>IH</sub> | Input high voltage<br>输入高电压          | 2.0V     | VCC+0.3V |
| V <sub>IL</sub> | Input low voltage<br>输入低电压           | -0.3V    | 0.8V     |
| V <sub>OH</sub> | High-level output voltage<br>高电平输出电压 | VCC-0.4V | --       |
| V <sub>OL</sub> | Low-level output voltage<br>低电平输出电压  | --       | 0.41V    |
| I <sub>OH</sub> | Sourcing current<br>拉电流              |          | 8mA      |
| I <sub>OL</sub> | Sinking current<br>灌电流               |          | 8mA      |

## 2. Can withstand Voltage Range / 能承受的电压范围

The signal with the maximum voltage range from -0.3V to 3.6V is as follows: COM1/2/3(TX&RX), SPI, LED\_STAT/RTK, PV, RTS\_SYS, PPS, EVENT1 and EVENT2.

所能承受电压的最大值范围是-0.3V~3.6V的信号如下：COM1/2/3（TX&RX），SPI，LED\_STAT/RTK，PV，RTS\_SYS，PPS，EVENT1和EVENT2。

## 3. Supply Voltage / 供电电压

VIN main power supply, voltage range: 3.3V~3.45V (DC), voltage ripple and spikes require less than 50mV.  
K807: LNA\_PWR, voltage range: 3.3V ~ 5.5V (DC), voltage ripple and spikes require less than 50mV.

VIN主供电电源，电压范围：3.3V~3.45V（直流），电压纹波和尖峰脉冲要求小于50mV。K807: LNA\_PWR，电压范围：3.3V~5.5V（直流），电压纹波和尖峰脉冲要求小于50mV。

#### 4. Add Surge Protection / 增加浪涌保护

In order to effectively protect against lightning strikes and surges, and prevent damage to the feed current limiting chip inside the module, it is recommended that users supply power to the antenna from outside the module and increase the surge protection function.

If you need to feed the antenna from the outside, it is recommended to use a high-voltage, high-power feeder chip; or add high-power protection devices such as gas discharge tubes, varistors, and TVS tubes to the feeder circuit.

为了有效防雷击、防浪涌，防止模块内部的馈电限流芯片损坏，建议用户从模块外部给天线供电并增加浪涌保护功能。

如需从外部为天线馈电，建议选用高耐压、大功率的馈电芯片；或在馈电电路上增设气体放电管、压敏电阻、TVS管等大功率的防护器件。

#### 5. Hardware Integration Considerations / 硬件集成注意事项

- 1) VCC power-on has a good monotonicity, and the initial level is lower than 0.4V, and the upstroke and ringing are guaranteed within the range of 5% VCC;
- 2) Use the VCC pin to provide a reliable power supply and all GND pins of the module are grounded;
- 3) The LNA\_PWR pin is connected to +3.3~5.5V voltage;
- 4) The module reset pin RTS\_SYS is for restoring the factory settings of the module. Please connect it correctly to ensure that the module can be reset reliably;
- 5) Special attention should be paid to:

Power supply: The guarantee of stable and low ripple power supply, the peak-to-peak value of the ripple voltage is preferably not higher than 50mVpp. It is recommended to use a power chip with a current output capacity greater than 2A to supply power to the module.

In addition to using LDO to ensure the power supply is pure, it is also necessary to consider:

- ①. Widening the power traces or using split copper surfaces to transmit current;
  - ②. Place the LDO as close to the module as possible in the layout;
  - ③. Avoid the power traces passing through high power with high inductance devices such as magnetic coils.
- 6) Antenna interface: The antenna line is as short and smooth as possible, avoiding acute angles; pay attention to 50Ω impedance matching;
  - 7) Avoid wiring directly under K807;
  - 8) The module is as far away from the high temperature airflow as possible.

- 1) VCC上电具有良好的单调性，且起始电平低于0.4V，上冲与振铃保障在5%VCC范围内；
- 2) 用VCC引脚提供可靠的电源且模块所有GND引脚接地；

- 3) LNA\_PWR脚接入+3.3~5.5V电压;
- 4) 模块复位引脚RTS\_SYS为恢复模块出厂设置, 请正确连接以保证模块可以可靠复位;
- 5) 在设计中应特别注意:  
供电: 稳定及低纹波电源的保证, 纹波电压峰峰值最好不高于50mVpp。建议采用电流输出能力大于2A的电源芯片给模块供电。  
除了可采用LDO保证供电纯净外, 还需要考虑:
  - ①. 加宽电源走线或采用分割铺铜面来传输电流;
  - ②. 布局上尽量将LDO靠近模块放置;
  - ③. 电源走线避免经过大功率与高感抗器件如磁性线圈。
- 6) 天线接口: 天线线路尽量短且顺畅, 避免走锐角; 注意50Ω阻抗匹配;
- 7) 避免在K807正下方走线;
- 8) 模块尽量远离高温气流。

## 6. Static Electricity Protection / 静电保护

Some components on K807 module are easily damaged by static electricity, which in turn affects the IC circuit and other components. Therefore, you should pay attention to electrostatic protection measures when using it.

- 1) When handling the module, try to wear gloves or finger cots and an anti-static wrist strap that meets the electrostatic protection standards;
- 2) During the process of taking the module, only the edge of the board should be taken, and the solder joints, circuit parts or components should not be directly touched to avoid sweat fingerprints from contaminating the solder joints;
- 3) The module should be protected by a soft protective pad between the module and the module interval during transportation;
- 4) When the module is idle, it should be placed on a soft protective pad (such as anti-static sponge pad, etc.), and do not stack at will;
- 5) The modules should be placed neatly and orderly, with a certain interval between modules to avoid collision with each other;
- 6) The module should be handled with care during use to prevent the module from being damaged by rough operation;
- 7) When powering on, pay attention to the positive and negative poles of the power supply and the voltage to avoid reverse connection and excessive voltage from burning the module;

K807模块上的部分元器件易受静电影响而损坏, 进而影响IC电路及其他元件。因此在使用时应注意做好静电保护措施。

- 1) 拿取模块时应尽量戴好手套或者指套以及符合静电防护标准的防静电腕带;
- 2) 模块拿取过程中应只拿取板卡的边缘部位, 不能直接接触焊点, 线路部分或者元器件, 避免汗液指印污染焊点;
- 3) 模块在运输过程中模块与模块间隔之间应该采用软性防护垫进行保护;
- 4) 模块闲置时应放置在软性防护垫上(如防静电海绵垫等), 不要随意堆叠;
- 5) 模块摆放应摆放整齐有序, 模块之间保持一定间隔, 避免相互碰撞;
- 6) 模块在使用过程中应该轻拿轻放, 防止粗暴作业损坏模块;
- 7) 上电时, 注意电源正负极以及电压, 避免反接和电压过高烧毁模块;

## 4. Application Connection Example / 应用连接示例

In this section, an application connection example of K807 OEM Module is presented via specific schematic diagrams. Per the instruction of these diagrams, you could easily build the communication circuits between K807 OEM Module and other terminals such as PC, GPRS or Bluetooth module, and some other devices with an UART.

本部分以具体电路的形式提供一个K807模块应用连接示例。参照下面的图示，您可以很方便建立K807模块和其他终端（如PC，GPRS模块，蓝牙模块或其他带有UART的设备）之间的通讯电路。

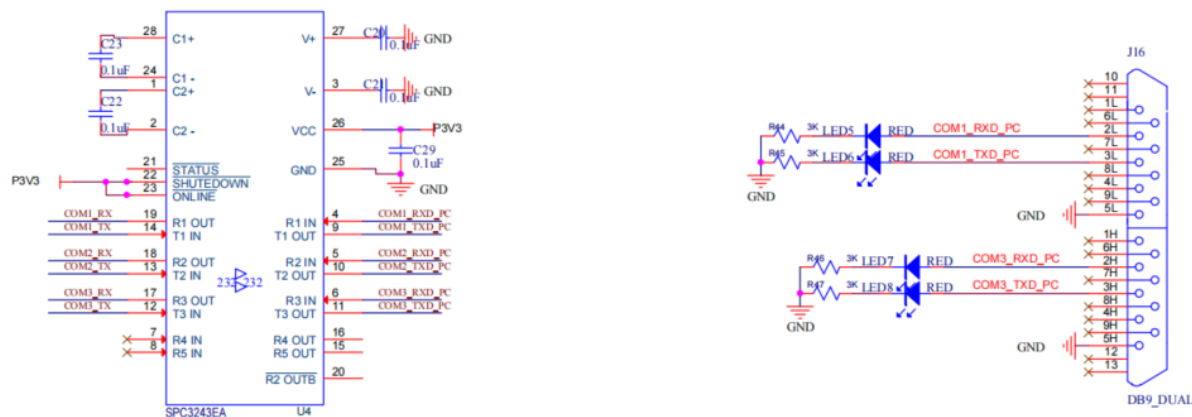


Figure 4. Connections between RS232 COM1, 2, 3 of K807 and Some Other Devices with An UART / K807 RS232 COM1、2、3与其他使用UART接口的设备之间的连接示意

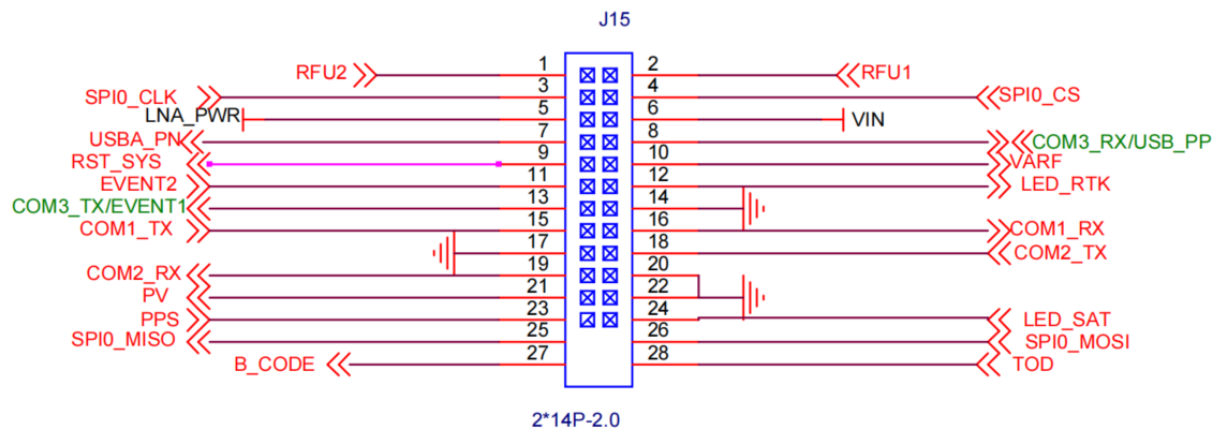


Figure 5. Minimum Hardware Design of K807 / K807 最小硬件设计



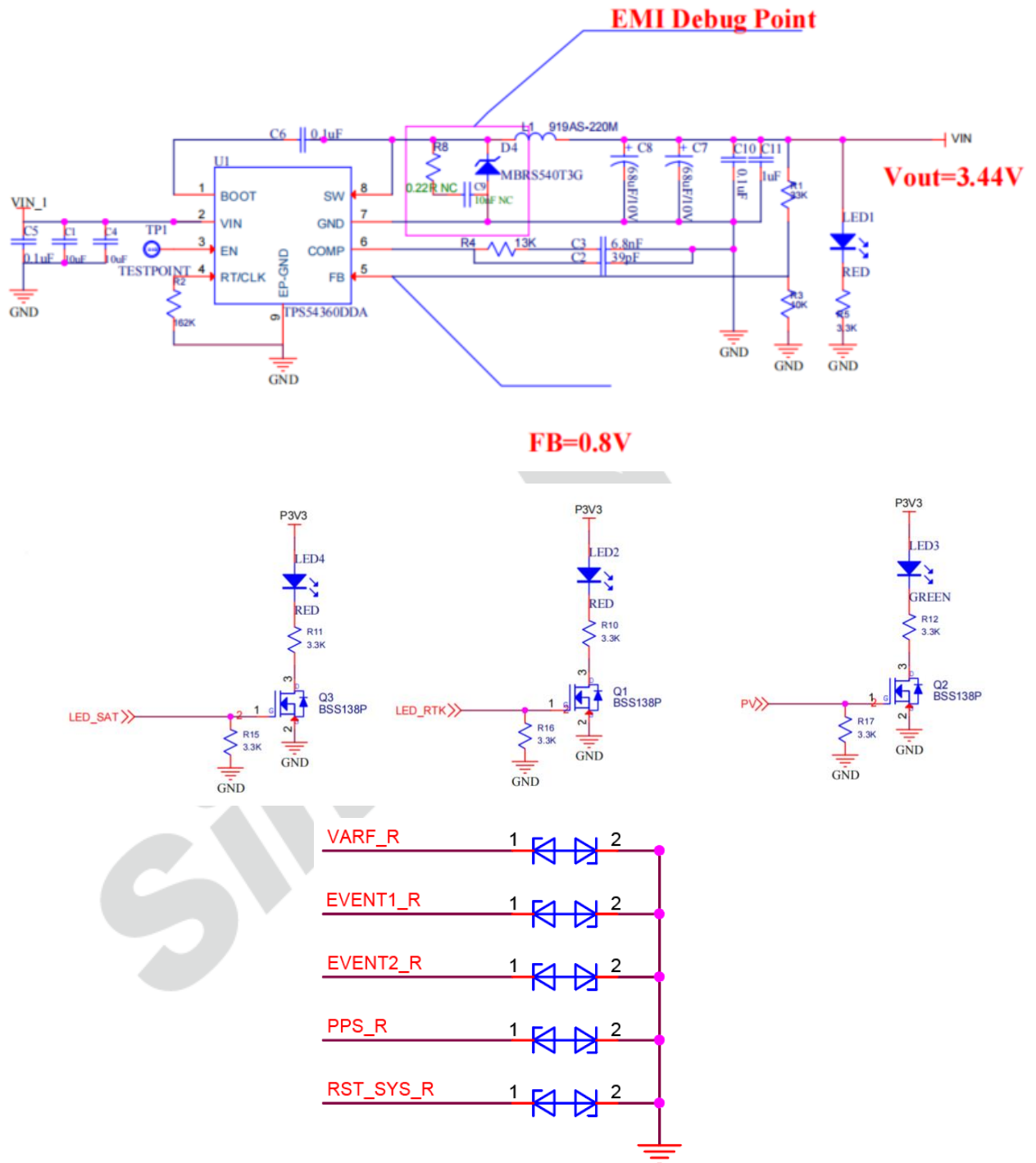


Figure 6. Recommend Design of K807 / K807 推荐硬件设计

## 5. Package / 包装

K807 boards are packed in anti-static foam, and 30 pieces of K807 boards are provided in each box.

K807板卡使用防静电泡棉方式包装，每盒提供30片K807板卡。

Table 5. K807 Package Description / K807 包装说明

| Project                 | Description                                            |
|-------------------------|--------------------------------------------------------|
| Board Packaging<br>模块包装 | Number of boxes:1 Pcs/Box<br>包装盒数量：1 个/盒               |
|                         | Number of anti-static foams:1 Pcs/Box<br>防静电泡棉数量：1 个/盒 |
|                         | Board quantity: 30 Slice/Box<br>板卡数量：30 片/盒            |

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